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Roll No

EC-6001-CBGS

B.E. VI Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Cellular Mobile Communication

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Draw the schematic diagram of a basic cellular system and describe the function of each unit.
b) Explain co-channel interference and how it affects the system capacity? Derive expression for signal to interference in terms of cluster size.
2. a) What is the primary purpose of cell splitting? Explain the cell splitting techniques.
b) Discuss the propagation between two fixed station over water or flat open-land.
3. a) Determine the path loss from point-to-point prediction model.
b) Discuss the gain and pattern relationship of mobile antenna.
4. a) Explain the propagation of signals over water or flat open area.
b) Explain how to obtain mobile point-to-point model.

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5.
 - a) Explain co-channel interference. Discuss real time co-channel interference measurement at mobile radio transceivers.
 - b) Explain briefly about types of Non co-channel interference.
6. Discuss in detail about frequency management and channel assignment.
7.
 - a) In a mobile radio environment, for hilly terrain of height $H = 100\text{m}$, the average cell site antenna height is about 50m , the mobile antenna height is about 3m and the communication path length is 5 km . Find the incident angle and the slope angle.
 - b) Explain the modulation process and characteristics for reverse CDMA channel.
8. Write short notes on (any two):
 - a) Cell Splitting
 - b) Mobile Antennas
 - c) TDD System
 - d) Mobile to Mobile Propagation
